Joshua Green

Electrical Engineer

joshuamdgreen@outlook.com
(416) 220-1799
London, Ontario
joshuagreen.ca
LinkedIn
GitHub

Education

Ph.D.
Electrical Engineering
Communication Systems and Data Networking
Network Security
Western University
Sept 2021 - Aug 2026
London, ON.

B.E.Sc
Electrical Engineering
Wireless Communication
Western University
Sept 2016 - Apr 2021
London, ON.

Skills

- **OOP Programming**
  - (C++, Python, MATLAB)
- **Python: Machine Learning**
  - (Pandas, Keras, TensorFlow, Numpy, more)
- **Other Programming**
  - (C, Assesmbly, Arduino)
- **Markup/Styling**
  - (Latex, HTML, CSS, XML)
- **Source Control**
  - (git)
- **Scripting**
  - (Linux, Bash)
- **PCB Circuit Design**
  - (Microcap, EAGLE, Soldering)
- **Allen Bradley PLC programming**
  - (FactoryTalk, DNET,RSLogix, RSLinx)
- **Engineering tools**
  - (AutoCAD, QuartusII, SOLIDWORKS)
- **Welding**
  - (MIG, Stick, Oxy-Acetylene)
- **Manufacturing**
  - (Lathe, Machining, Plasma-cutting)

Publications and Awards

- IEEE PIMRC Conference Paper Publication: "Shared Experiential History for Encryption Based Peer-To-Peer Authentication"
- 2023 Graduate Symposium Outstanding Presentation in Communication Systems Engineering
- Dean's List Honoree 2017-2018

Projects

- Lung disease classification using CNN and Transfer learning
- Capstone: Smart cane elderly alert system
  - Capstone role: GPS data polling, data formatting, SMS message formatting, send SMS to contacts
- Project 2: Welder Power Supply with PID
- Project 1: Student card automatic door access

Work Experience

Graduate Research and Teaching Assistant
Western University - London, ON Sept 2021 - Current

- Drive pioneering research in wireless network security.
- Engineer novel methods for Authentication and Trust Assessment
- Monitor and mark labs, midterm and final exams, and assignments for the following courses:
  - ECE4416/4415 (ECE Design Project Course)
    - Guide students through their final engineering project.
  - ECE3331B (Introduction to Signal Processing)
    - Course is heavily reliant on MATLAB and Simulink.
  - ECE4436A (Networking: Principles, Protocols and Architectures)
    - Linux · Socket Programming · Virtual Machines · Python
  - ECE2236B (Magnetic Circuits and Transmission Lines)
    - Electromagentic Theory

Electrical Engineering Intern (Controls)
General Motors - Ingersoll, ON May 2019 - May 2020

- Create original PLC program solutions using Allen Bradley PLC Software (Factory Talk, DNET, RSLogix, RSLinx),
- Departmental DNET backup during the plant shutdown down due to first covid-19 wave,
- Update electrical drawings in AutoCAD and document the changes in the company database,
- Cross-functional teamwork at a large scale,
- Quality defect isolation and solutions,
- Project Management (PM delegation, SAP ordering, supervising engineering projects).
- Upholding Ontario Electrical Safety Code (OESC).
- Update company database for: Lockout Zones/Tags, Safety zones, PLC programs, DNET.